

A Hybrid Matrix Completion + Liquid Neural Network Framework for Groundwater Level Imputation

Coupling Spatial Correlation Structure with Continuous-Time Temporal Dynamics for Long-Gap Reconstruction

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1 The Problem

Public groundwater monitoring networks (e.g., USGS NWIS) contain **millions of measurements** but are plagued by **long temporal gaps** spanning months to years. These gaps prevent basin-scale storage estimation critical for drought response and water policy.

592

GSLB Wells

70%

Best Coverage

288

Months (2000-2023)

Prior methods (ELM, LSTM, GRU) treat each well independently. Matrix completion has been applied only once to groundwater (Sharma et al., 2024) as a standalone method. No prior work couples spatial MC with temporal neural dynamics.

2 MC+LNN Pipeline

Stage 1: PCHIP

Fill gaps ≤ 24 months
Shape-preserving
polynomial interpolation

Stage 2a: ARCHI

Top-15 correlated donor
wells via OLS regression
Trend-aware initialization

Stage 2b: MC

Softimpute SVD on
composite matrix
donors + GLDAS + sin/cos

Stage 2c: LNN

CFC reservoir dynamics
MC as input placeholder
Trained on observations only

Key insight: MC provides **spatial context** (cross-well correlations), LNN provides **temporal dynamics** (nonlinear continuous-time evolution). MC predictions drive the LNN reservoir, but readout trains only on real observations.

$$\text{LNN CFC: } x(t+dt) = x \cdot e^{-\text{leak} \cdot dt} + (b / \text{leak}) \cdot (1 - e^{-\text{leak} \cdot dt}) \quad \text{where } b = \tanh(W_{in} \cdot \text{input} + W_{res} \cdot x)$$

3 What's New

First framework to jointly exploit:

Spatial Structure

Matrix Completion exploits correlation across the full 592-well network via low-rank SVD with GLDAS auxiliary rows and seasonal encoding

Temporal Dynamics

Liquid Neural Network with closed-form continuous-time cells (Hasani et al., 2022) captures nonlinear drought response, recharge lags, and seasonal patterns

vs. Prior approaches:

- ELM (Evans et al., 2020): Temporal only, no cross-well structure
- LSTM/GRU (Jeong et al., 2020; Wunsch et al., 2021): Each well independent
- Softimpute (Sharma et al., 2024): Spatial only, no temporal model
- MC+LNN (ours): Spatial + Temporal, coupled

4 Cross-Validation Results (592 wells, Great Salt Lake Basin)

A. Random Missing Data (50 trials each)

% Removed	KGE	R ²	RMSE (ft)	MAE (ft)
5%	0.837 ± 0.085	0.771	2.53	1.98
10%	0.843 ± 0.077	0.770	2.70	2.06
20%	0.853 ± 0.051	0.787	2.64	1.98
30%	0.844 ± 0.048	0.770	2.79	2.03
40%	0.851 ± 0.030	0.784	2.75	2.03
50%	0.847 ± 0.035	0.788	2.74	2.03

KGE remains >0.84 even when 50% of data is removed.

B. Consecutive Year Gaps (20 trials each)

Gap Size	KGE	R ²	RMSE (ft)	MAE (ft)
1 year	0.783 ± 0.116	0.703	2.98	2.28
2 years	0.792 ± 0.081	0.721	2.97	2.29
3 years	0.802 ± 0.076	0.730	3.02	2.33
4 years	0.806 ± 0.055	0.738	2.96	2.15
5 years	0.815 ± 0.063	0.744	2.91	2.20

KGE >0.78 even for 5-year consecutive gaps.

5 PCHIP Matters

Replacing LNN-based small-gap fill with PCHIP in Stage 1 improves performance across the board:

Random Missing (avg KGE)



Consecutive Gaps (avg KGE)



+10%

Random KGE

+19%

Consec. KGE

PCHIP densification improves donor correlations for the MC stage, cascading through the pipeline.

6 Each Stage Adds Value

Ablation study isolating each component:

Method	KGE (random)	KGE (consec.)
MC alone	0.24	-0.07
LNN alone	0.65	0.45
MC + LNN	0.85	0.80

MC provides spatial placeholders. LNN refines them with temporal dynamics. **Neither alone achieves what the coupled framework delivers.**

7 Study Area: Great Salt Lake Basin

- 93,000 km² closed hydrologic system, 4 states
- 592 eligible wells with ≥10 observations (2000-2023)
- Concentrated pumping along the **Wasatch Front**
- WTE range: **4,200 - 7,200 ft**
- GLDAS soil moisture (5 temporal scales) as auxiliary input
- Framework imputed **all 592 wells to 100% completeness**

Data Summary

Wells: 592
Months: 288 (Jan 2000 - Dec 2023)
Aux variables: 5 (GLDAS soil moisture)
Donor wells: top 15 by correlation
MC ranks tested: 3, 5, 8, 10, 12
LNN ensemble: 3 models
Optimization: 8 trials/well

8 Conclusions

- MC+LNN achieves KGE > 0.84 under random missing data from 5-50%, with remarkable consistency
- KGE > 0.78 for 1-5 year consecutive gaps -- the hardest imputation scenario
- PCHIP for small gaps is critical: +10-19% KGE over LNN-based small-gap filling
- The coupling is essential: MC alone (KGE 0.24) and LNN alone (0.65) both underperform the integrated pipeline (0.85)
- Successfully imputed all 592 GSLB wells to temporal completeness

Next: Spatiotemporal interpolation of the complete dataset to continuous groundwater-level fields for GRACE leakage correction (Paper 3)

